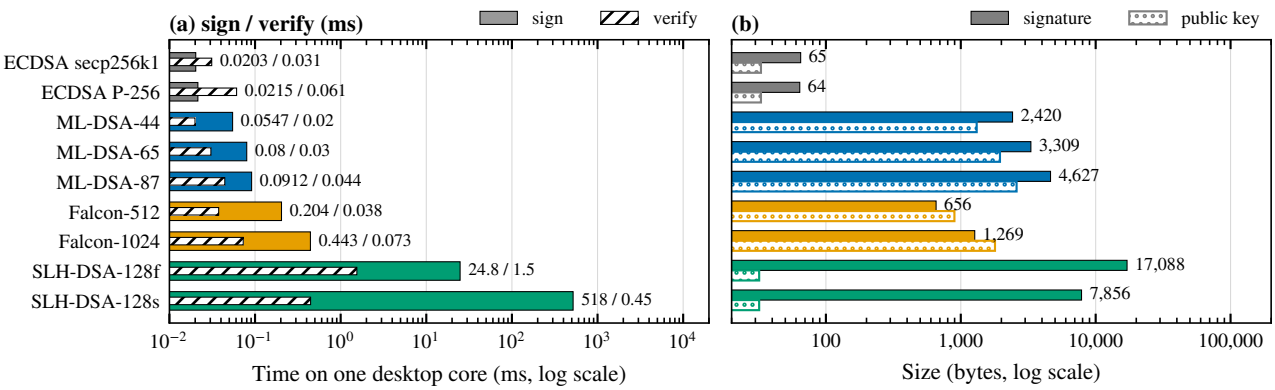

(a) sign / verify (ms)
sign
verify
ECDSA secp256k1
ECDSA P-256
ML-DSA-44
ML-DSA-65
ML-DSA-87
Falcon-512
Falcon-1024
SLH-DSA-128f
SLH-DSA-128s
0.0203 / 0.031
0.0215 / 0.061
0.0547 / 0.02
0.08 / 0.03
0.0912 / 0.044
0.204 / 0.038
0.443 / 0.073
24.8 / 1.5
518 / 0.45
10^{-2}
10^{-1}
10^{0}
10^{1}
10^{2}
10^{3}
10^{4}
Time on one desktop core (ms, log scale)
(b)
signature
public key
65
64
2,420
3,309
4,627
656
1,269
17,088
7,856
100
1,000
10,000
100,000
Size (bytes, log scale)